ML20 Build the project baseline

Week 7, Your project | Session 20 of 24 | Machine Learning | New library: sklearn, seaborn

Today's question. How much does a plain guess miss by, before any model helps?

Objective. By the end: you can split data into training, validation and locked test, fit a baseline on training only, and draw one descriptive chart.

Before this session: ML19 Project data and a safe plan. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
three-way split	Training, validation and final test parts.	Practice, mock and real exam.
baseline error	The average miss of the plain guess on validation.	How wrong the average guess is.
scatter plot	Each row is a dot; its place shows two values.	Dots on a map of distance and fare.
training only	Fit anything only on the training part.	Learning only from the textbook.

Block A. Split and fit the baseline

Use the fares as the worked example, then repeat with your own dataset. Lock 20% away, take 25% of the rest for validation, and fit the mean guess on training only.

Predict. Predict the three sizes and whether the baseline will miss fares by more than UGX 3000 on average. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.dummy import DummyRegressor
from sklearn.metrics import mean_absolute_error
fares = pd.read_csv("synthetic_fares.csv")
X = fares[["distance_km", "rush_hour", "raining"]]
y = fares["fare_ugx"]
X_dev, X_test, y_dev, y_test = train_test_split(X, y, test_size=0.2, random_state=7)
X_train, X_val, y_train, y_val = train_test_split(X_dev, y_dev, test_size=0.25,
    random_state=7)
baseline = DummyRegressor(strategy="mean").fit(X_train, y_train)
val_baseline_mae = mean_absolute_error(y_val, baseline.predict(X_val))
print(len(X_train), len(X_val), len(X_test), round(val_baseline_mae))
```

Run it, then compare. The cell prints 90, 30, 30 and 4361. The plain guess misses the validation fares by about UGX 4361 on average. Any model you build must beat that on the same validation trips. The baseline was fitted on the 90 training trips only. The 30 test trips stay locked.

Investigate.

1. Why is the baseline fitted on training only?
2. What would make the baseline number meaningless?

Modify. Print the share of the 150 trips used for training.

Block B. One descriptive chart

Plot the training trips: distance on one axis, fare on the other. Label both with units.

Predict. Predict the shape of the dots: rising, falling or no pattern. Write your guess on paper first.

Your guess, written before you run it

```
import seaborn as sns
import matplotlib.pyplot as plt
train = X_train.copy()
train["fare_ugx"] = y_train
sns.scatterplot(data=train, x="distance_km", y="fare_ugx")
plt.xlabel("Distance (km)")
plt.ylabel("Fare (UGX)")
plt.title("Fare against distance, training trips")
plt.show()
train_corr = train["distance_km"].corr(train["fare_ugx"])
```

Run it, then compare. The dots rise from left to right with a lot of scatter. The correlation on the 90 training trips is about 0.76. Longer trips cost more, but two trips of equal length can differ by thousands of shillings. That scatter is why even the best line will still miss.

Investigate.

1. What does the scatter around the rising pattern tell you?
2. Why plot only the training trips?

Modify. Colour the dots by rush_hour by adding hue="rush_hour" to the plot.

Make

Complete the three-way split, the baseline and one labelled chart for your own project dataset. Save the notebook, your data dictionary and the cleaning log. If no cleaning was needed, record the checks that found nothing.

Save your work as ML20. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
Baseline and model score exactly the same	Both scores were computed from the same predictions, so the baseline is not a separate guess.	Build the baseline from the training mean alone and score the model with its own predict, both on X_val and y_val.
ValueError: Found input variables with inconsistent numbers of samples	X and y have different lengths because one was filtered and not the other.	Create X and y from the same table at the same step.

Exit ticket

The baseline error on validation is the number my model has to beat.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.